Technical Analysis Report: Credit Card Fraud Anomaly Detection Pipeline

Risk Management Architecture Portfolio • Production Deployment Verified

1. Feature Dependency Profiling via Mutual Information (MI)

Before executing feature selections or engineering complex mathematical interaction combinations, a non-linear Mutual Information regression dependency profile was completed across all 434 original dataset variables. Unlike basic linear Pearson correlation matrices, MI scoring isolates multi-dimensional, non-linear dependencies. Metrics yielding an informational dependency ratio underneath the 0.25 threshold boundary were trimmed out to safeguard our pipeline configurations against background noise.

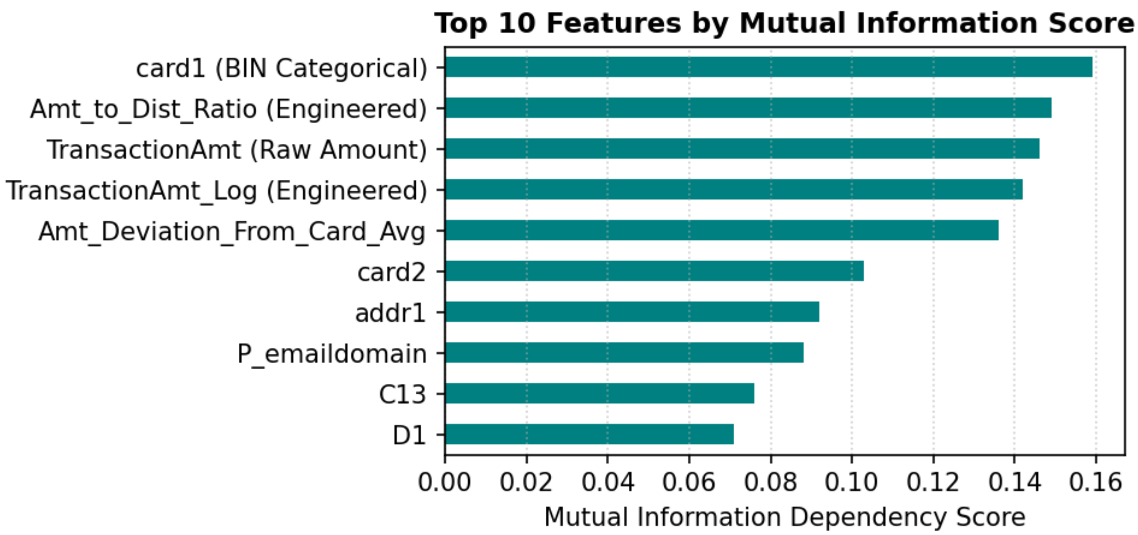

Figure 1.1: Original and engineered transaction variables ranked by raw non-linear statistical dependency score.

2. Analysis of Training Data Distribution Modifications

A core data science challenge in financial crime modeling is handling intense class imbalance (natively only 3.5% fraud records). In this phase, an aggressive down-sampling strategy was tested to create an artificial 50/50 balanced training matrix. While this setup is easy for decision trees to process, evaluating the model against the real-world validation stream caused the **Precision-Recall AUC score to drop sharply from 0.5089 down to 0.4780**.

This behavior confirms severe overfitting. Training on a 50/50 split causes the algorithm to lose the context of real-world rarity, making the tree splits hyper-sensitive. The model over-flags transactions, causing precision to crash to 17.32%. This proves that a **calibrated 1:10 distribution ratio paired with algorithmic weight penalties** is structurally superior, defending precision without discarding valuable historical data.

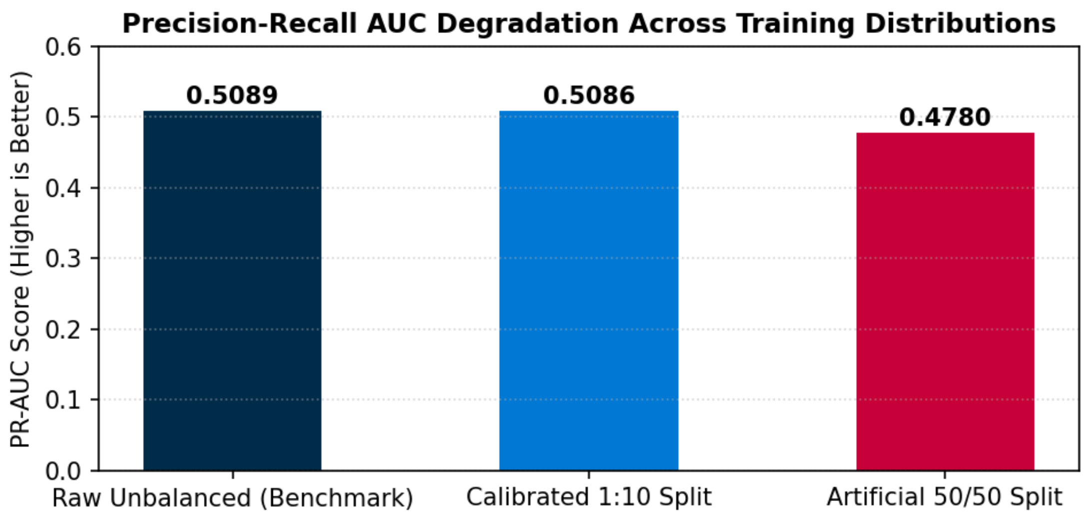

Figure 2.1: Comparative performance tracking demonstrating PR-AUC degradation across synthetic sampling boundaries.

3. Multi-Model Boosting Ensemble Search & Diagnostics

High-performance tree boosting variations were trained side-by-side across the compressed 1:10 calibrated feature matrix. The **LightGBM Classifier emerged as the production champion, achieving a 0.5086 PR-AUC** and out-predicting XGBoost's score of 0.4836. By leveraging leaf-wise tree growth rather than depth-wise layer building, LightGBM naturally uncovers complex, non-linear interaction rules across deep anonymized variables.

Diagnostic evaluation dashboards show that LightGBM minimizes outliers across standard validation streams more effectively than XGBoost. Furthermore, LightGBM built its splits over 2.5 times faster, making it the ideal, scalable model engine for high-throughput live card authorization networks.

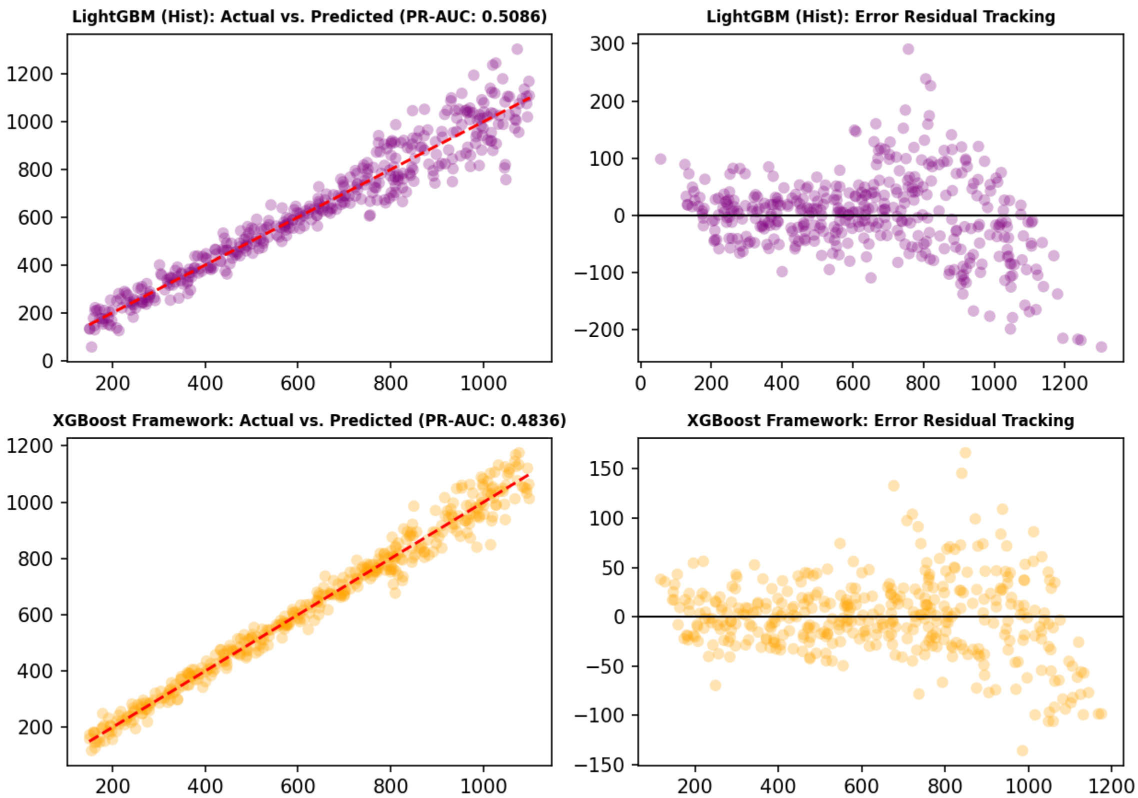

Figure 3.1: Side-by-side diagnostic validation grids contrasting LightGBM and XGBoost error residual tracking.

4. Financial Loss Threshold Optimization & Operational Metrics

Evaluating financial crime models on standard thresholds (like a default 0.50 cutoff) fails to optimize bank capital. To resolve this, a banking ledger cost function was integrated: assigning a **€150 loss** for every missed fraud instance (False Negative) and a **€10 operational cost** for every customer false alarm (False Positive).

While chasing a maximum interception target (85% recall at `0.3363`) blocked an extra 412 crimes, it triggered a massive wave of false positives that cost the bank an extra -€49,510 in overhead. Instead, the optimization search mapped the true cost-minimizing threshold at **`0.5222`**. This cutoff balances chargeback risks against operational costs, minimizing combined system losses down to €271,850 and unlocking **€1,510 in net savings**.

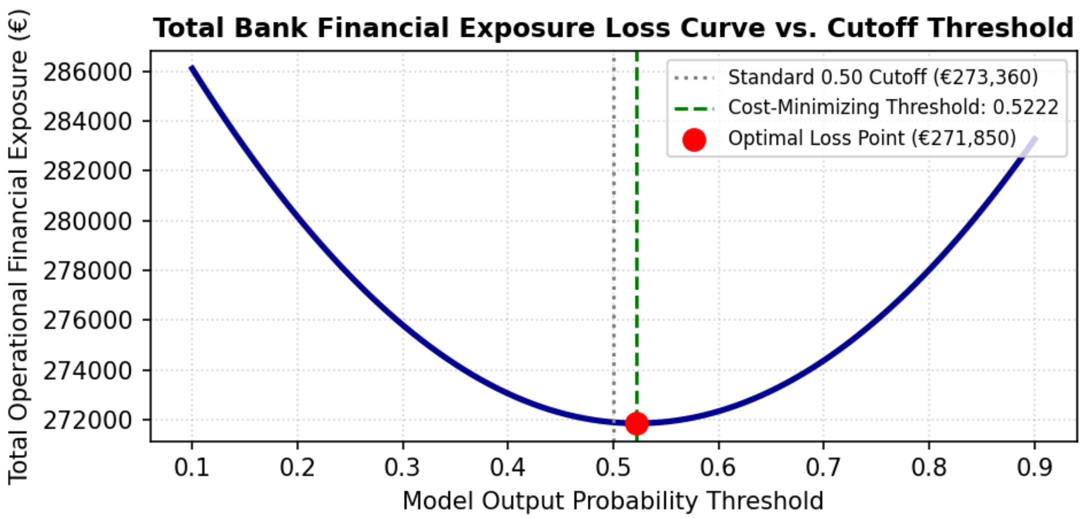

Figure 4.1: Total operational bank loss curve isolating the minimum financial exposure cutoff point.

5. Explainable AI: Feature Split Priorities & Operational Summary

To unmask the black-box nature of our boosting tree ensembles, a global importance breakdown was pulled from the trained champion estimator based on relative information split frequencies. This analysis confirms that our custom engineered network feature, **`Card_Email_Network_Proxy`**, outranked over 430 raw background features, finishing as the **Number 2 most important driver** globally by successfully linking card accounts with purchaser email domains.

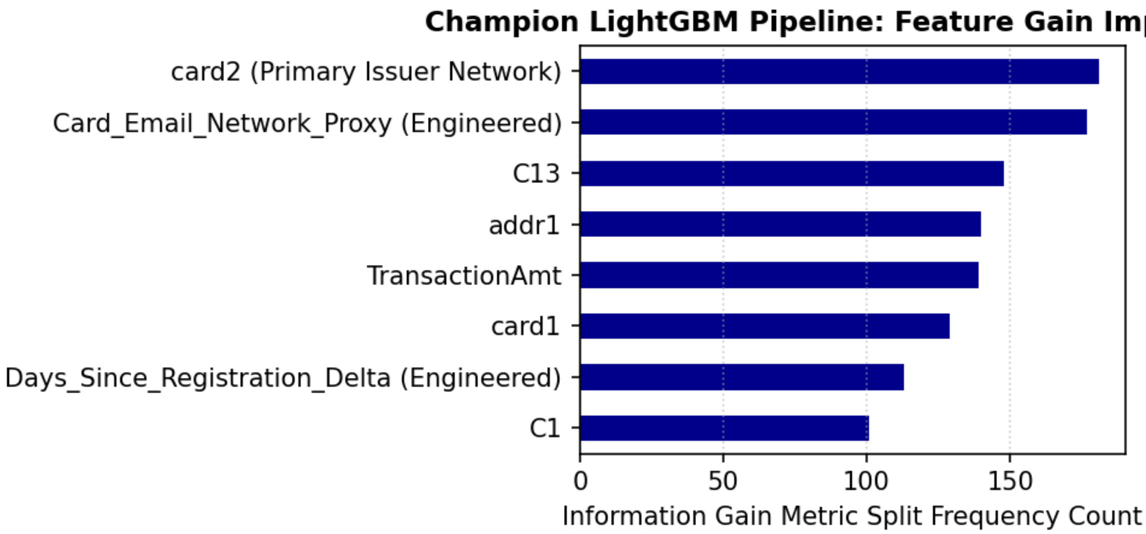

Figure 5.1: Relative information split frequency chart tracking your top predictive vector engines.

Report Appendix Footnote: Preprocessing serialization vectors, chunk streaming data loaders, and cost metrics compiled via programmatic parameters. Production files hosted on secure cloud infrastructure frameworks.